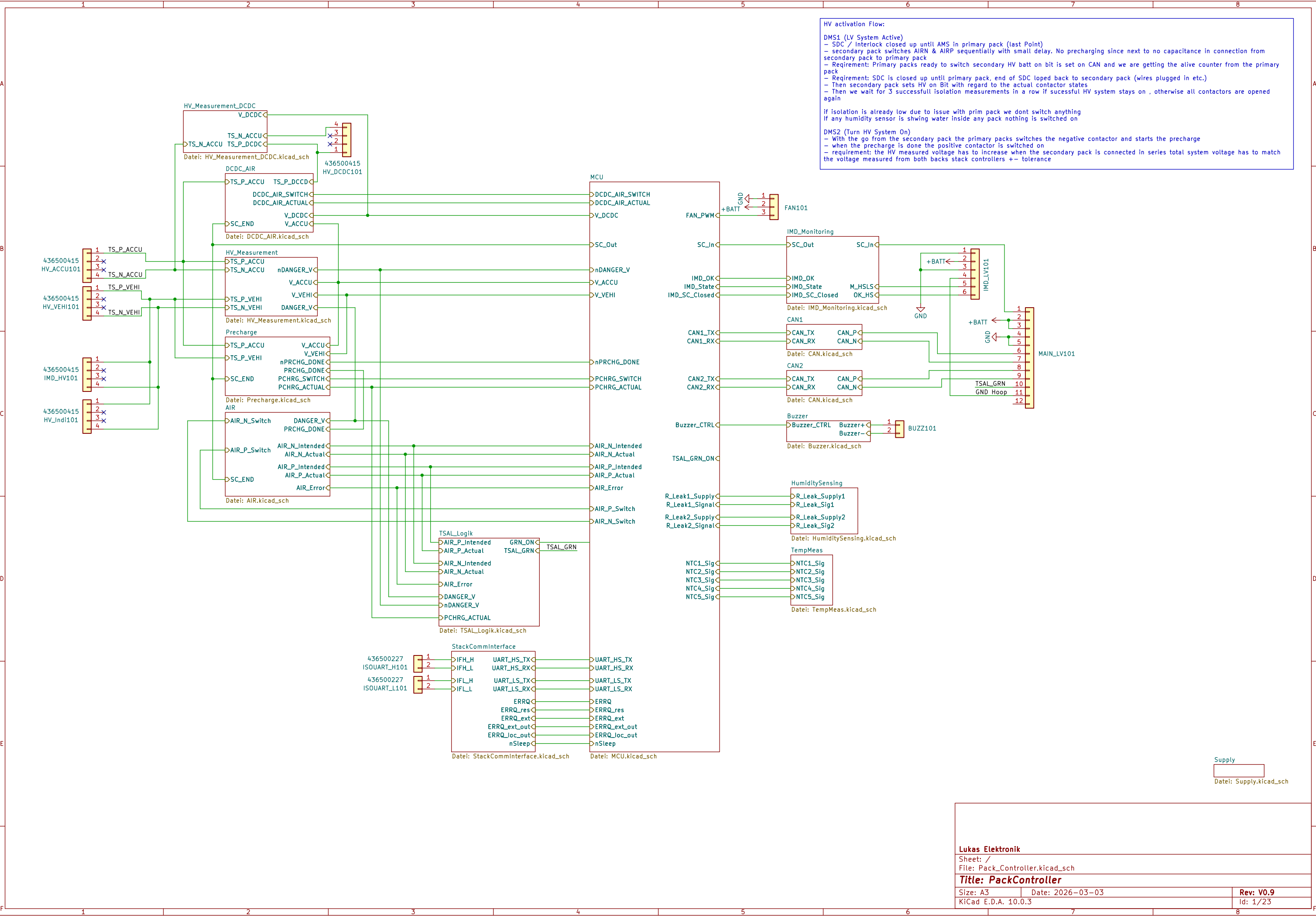




HV activation Flow:

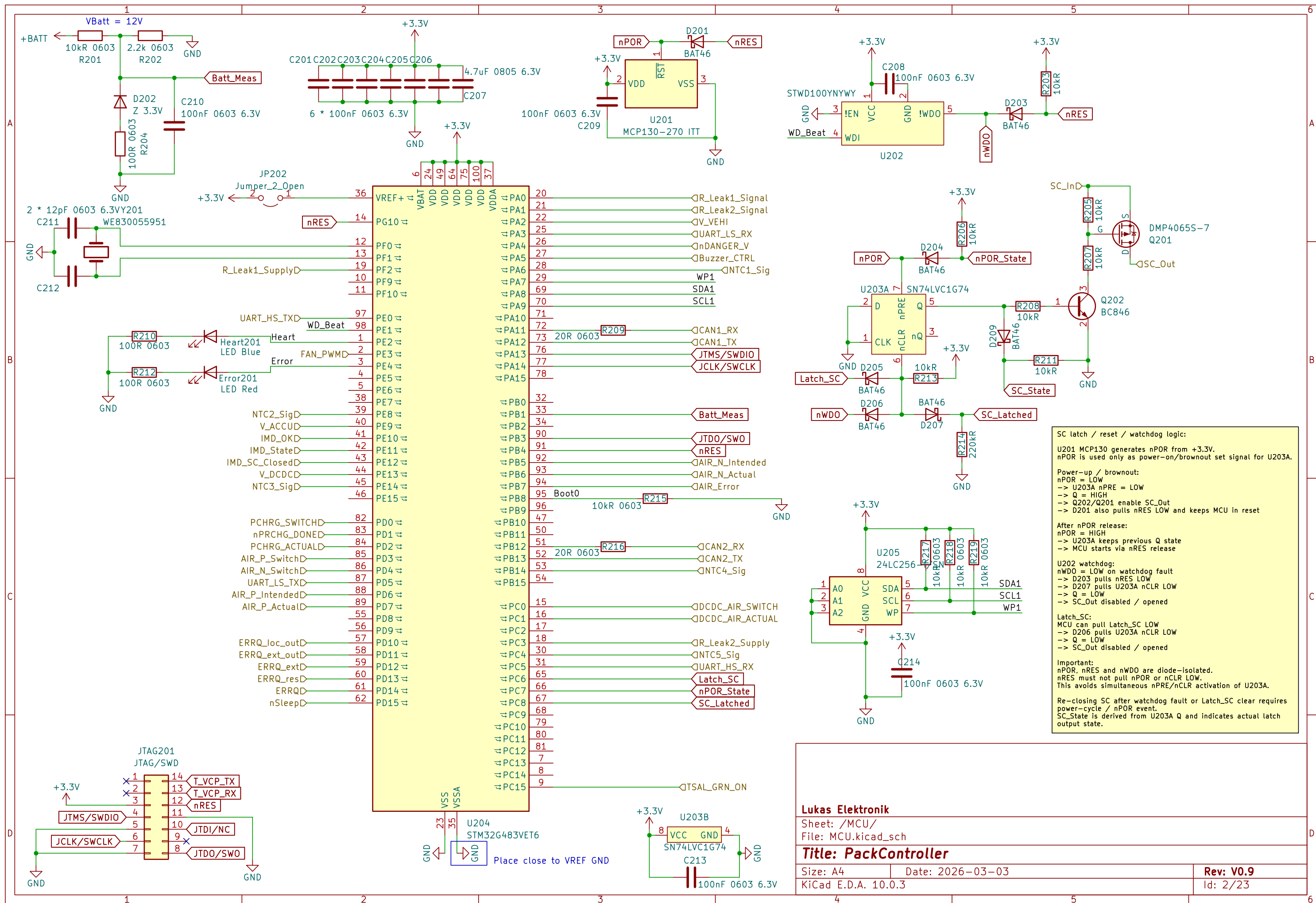
DMS1 (LV System Active)

- SDC / Interlock closed up until AMS in primary pack (last Point)
- secondary pack switches AIRN & AIRP sequentially with small delay. No precharging since next to no capacitance in connection from secondary pack to primary pack
- Requirement: Primary packs ready to switch secondary HV batt on bit is set on CAN and we are getting the alive counter from the primary pack
- Requirement: SDC is closed up until primary pack, end of SDC loped back to secondary pack (wires plugged in etc.)
- Then secondary pack sets HV on Bit with regard to the actual contactor states
- Then we wait for 3 successfull isolation measurements in a row if successful HV system stays on , otherwise all contactors are opened again

If isolation is already low due to issue with prim pack we dont switch anything
If any humidity sensor is shwing water inside any pack nothing is switched on

DMS2 (Turn HV System On)

- With the go from the secondary pack the primary packs switches the negative contactor and starts the precharge
- when the precharge is done the positive contactor is switched on
- requirement: the HV measured voltage has to increase when the secondary pack is connected in series total system voltage has to match the voltage measured from both backs stack controllers +- tolerance



Lukas Elektronik

Sheet: /MCU/

File: MCU.kicad_sch

Title: PackController

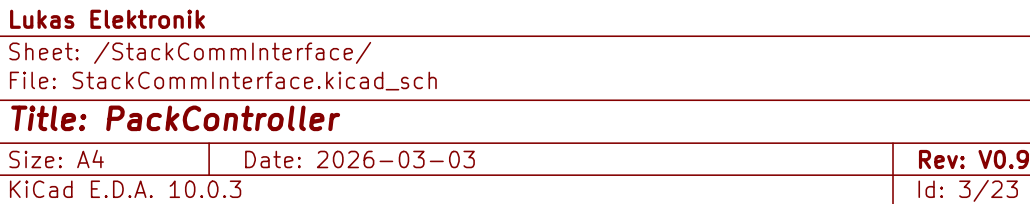
Size: A4

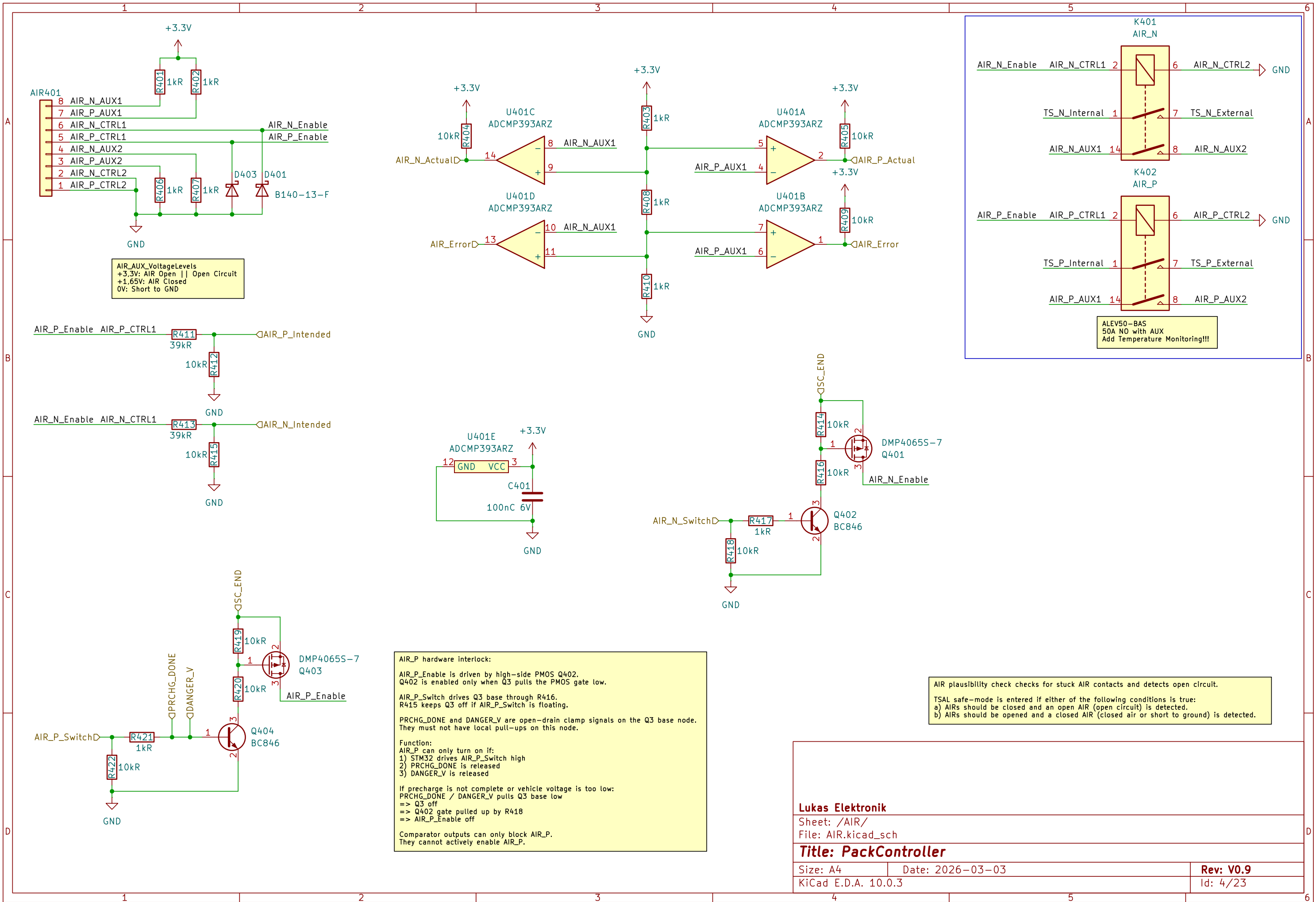
Date: 2026-03-03

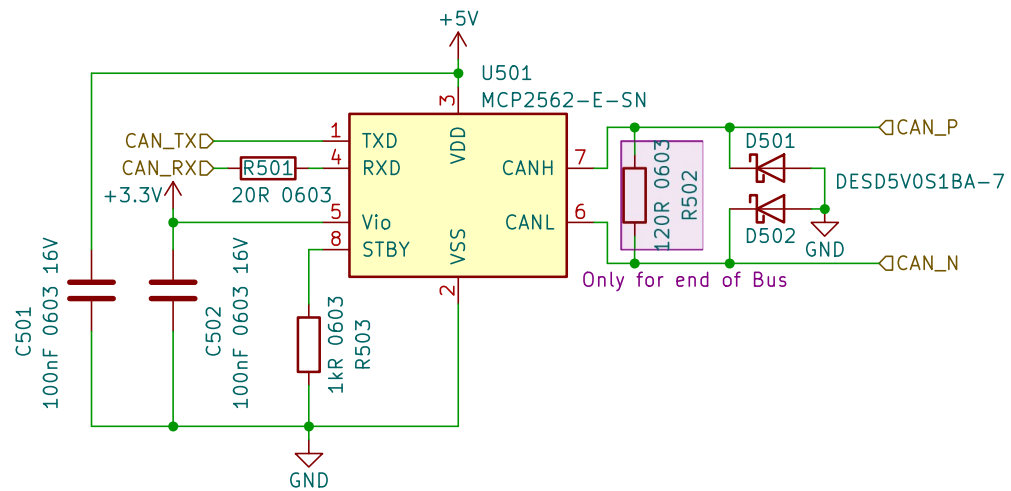
Rev: V0.9

KiCad E.D.A. 10.0.3

Id: 2/23







Lukas Elektronik

Sheet: /CAN1/
File: CAN.kicad_sch

Title: PackController

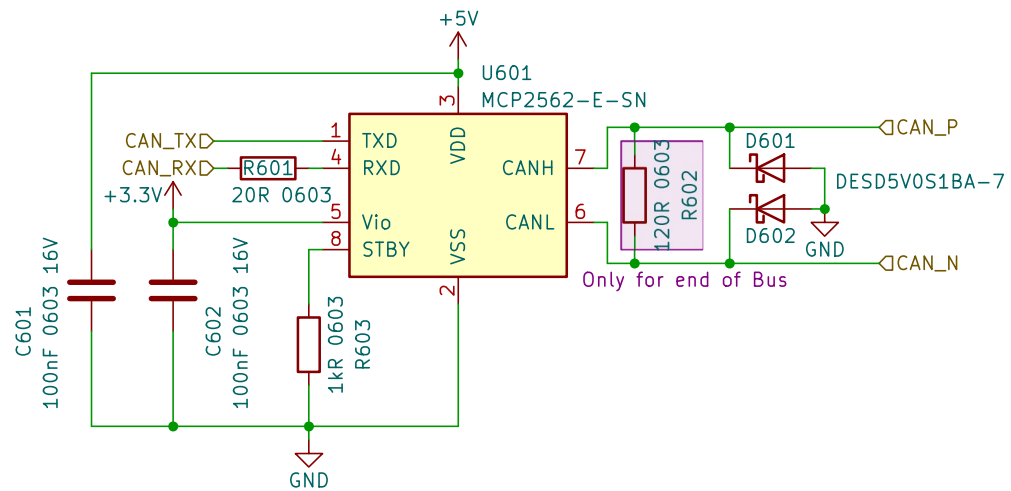
Size: A4

Date: 2026-03-03

Rev: V0.9

KiCad E.D.A. 10.0.3

Id: 5/23



Lukas Elektronik

Sheet: /CAN2/
File: CAN.kicad_sch

Title: PackController

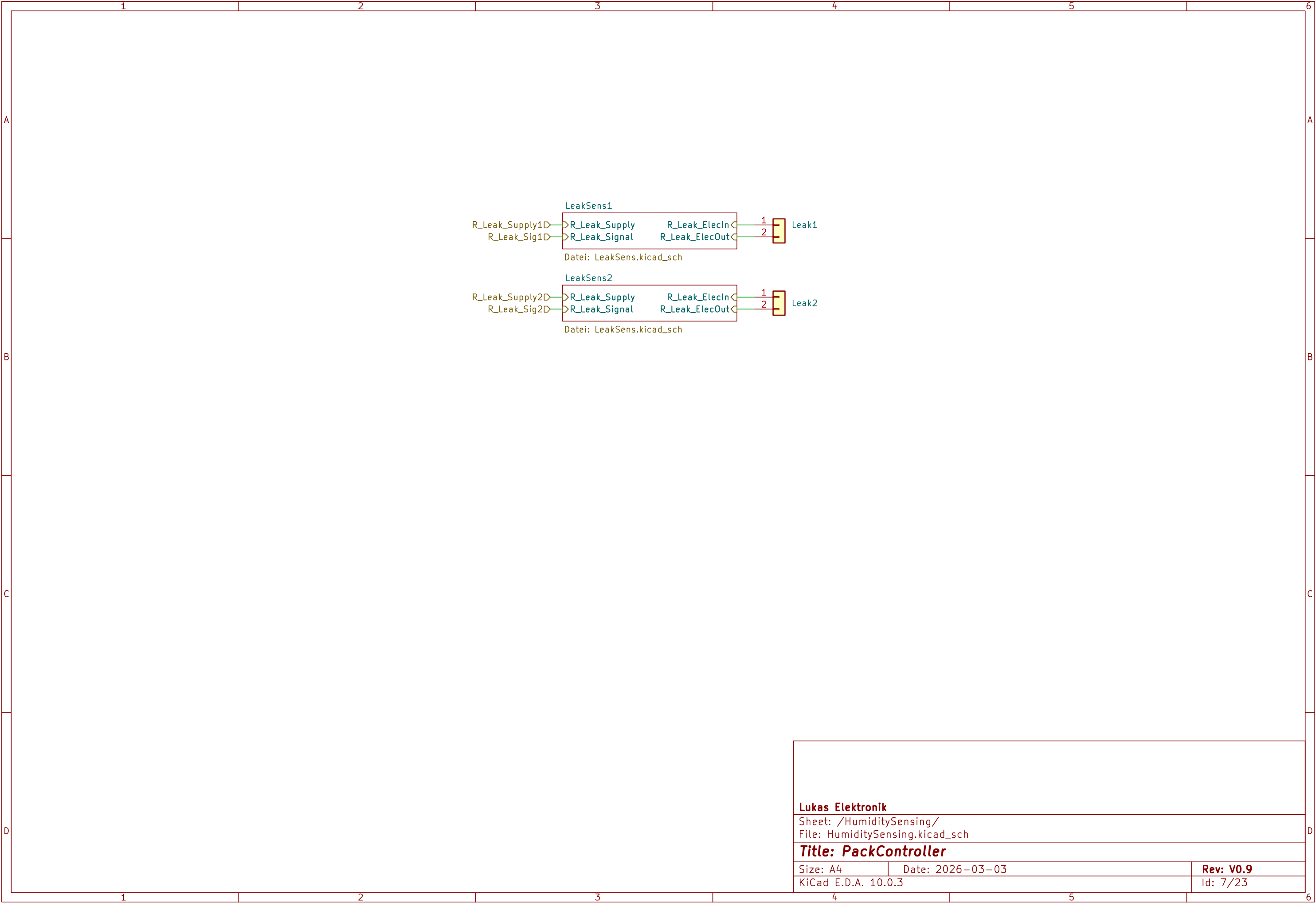
Size: A4

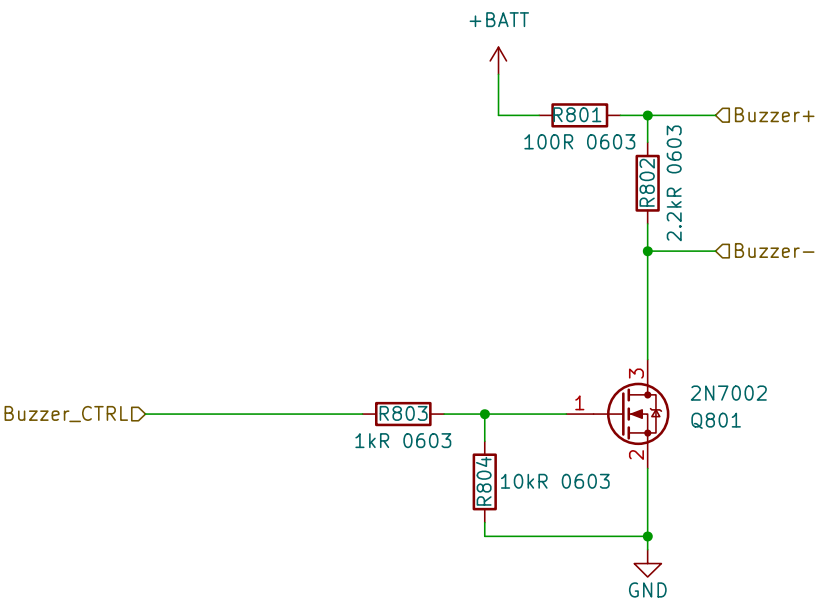
Date: 2026-03-03

Rev: V0.9

KiCad E.D.A. 10.0.3

Id: 6/23





Passive piezo buzzer driver.
Target part: CPT-2305-90PM, externally driven, 12 Vp-p, 4 kHz, typ. 24 nF.

Drive with STM32 PWM:
f_PWM ≈ 4 kHz
duty ≈ 50 %

Q15 is a low-side NMOS switch.
R37 discharges the piezo during MOSFET off-time:
 $\tau = R37 * C_{PIEZO} \approx 2.2k * 24nF \approx 53\mu s$.

R36 limits peak current and EMI.
No additional series coupling capacitor required.

Lukas Elektronik

Sheet: /Buzzer/
File: Buzzer.kicad_sch

Title: PackController

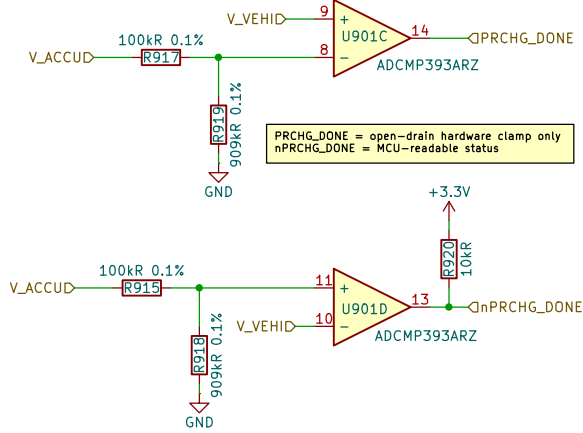
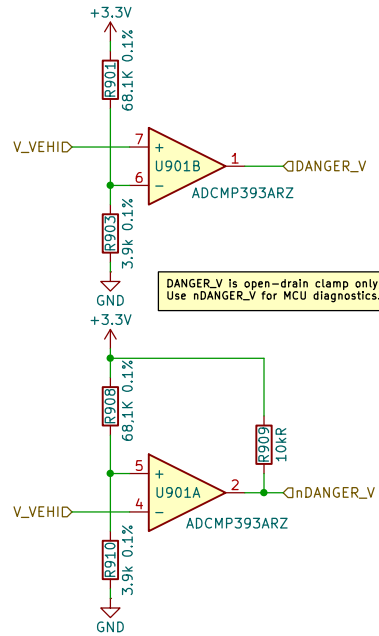
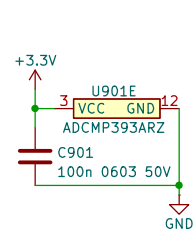
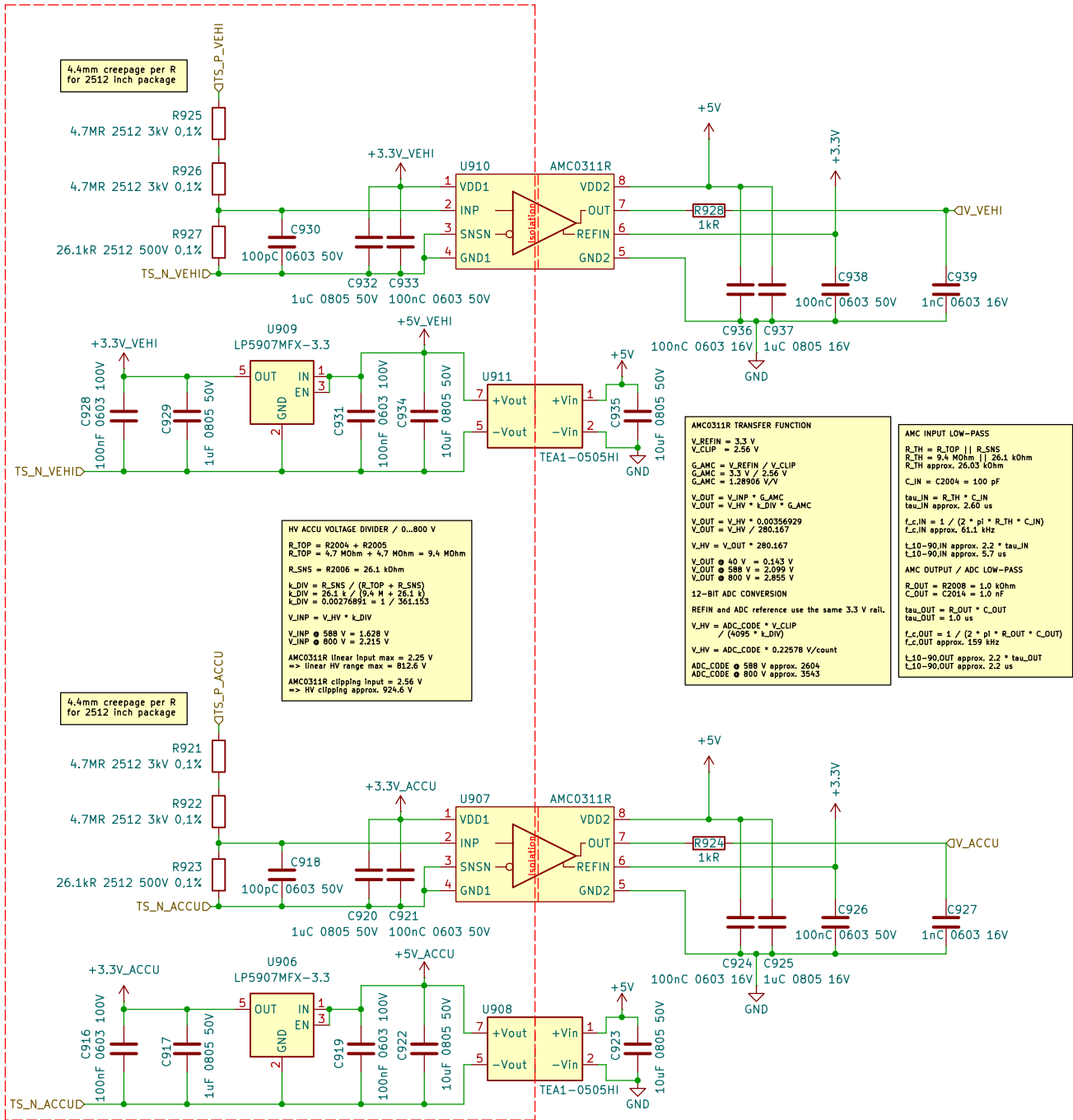
Size: A4

Date: 2026-03-03

Rev: V0.9

KiCad E.D.A. 10.0.3

Id: 8/23



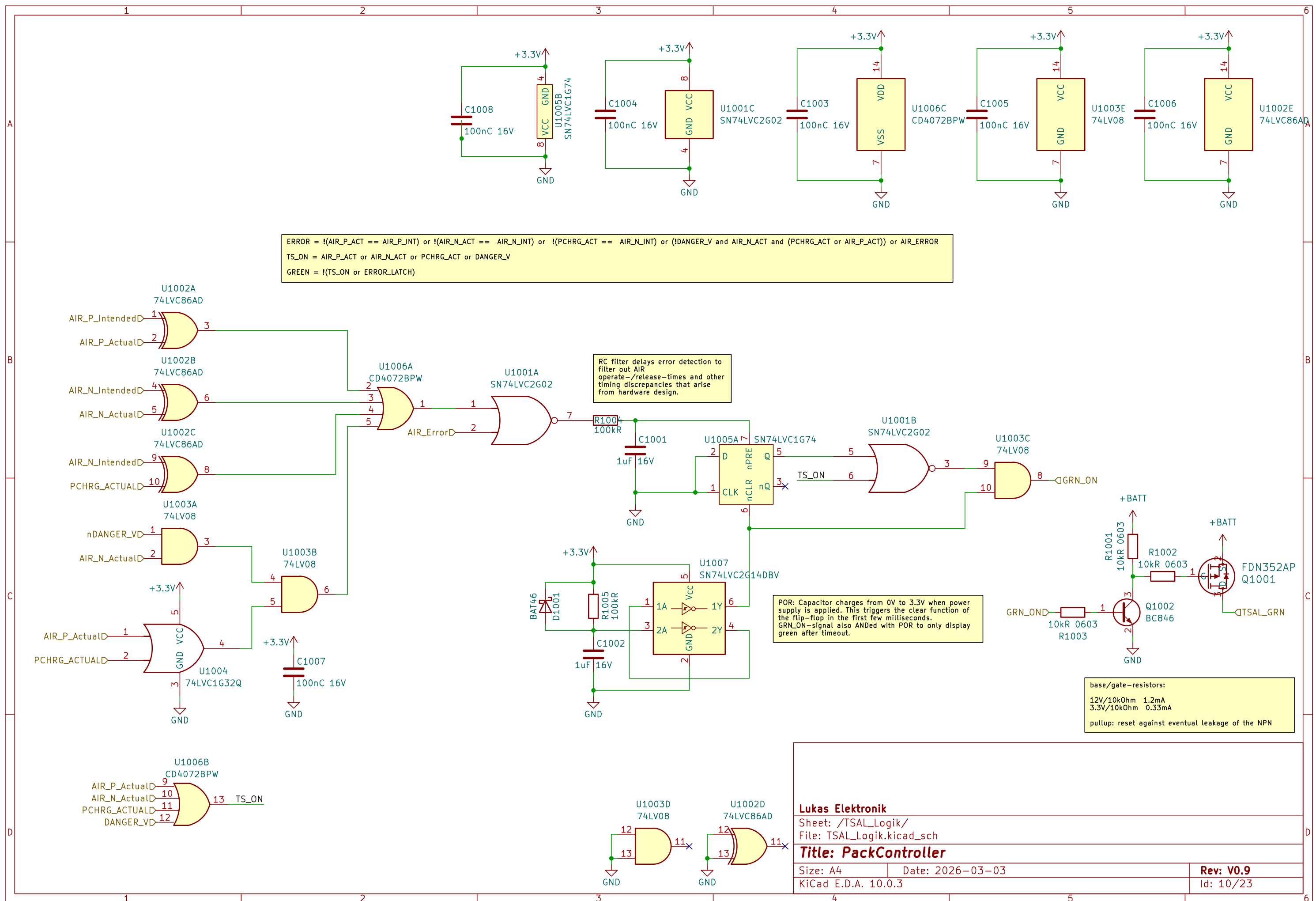
Lukas Elektronik

Sheet: /HV_Measurement/
 File: HV_Measurement.kicad_sch

Title: PackController

Size: A3 Date: 2026-03-03
 KiCad E.D.A. 10.0.3

Rev: V0.9
 Id: 9/23



Q1102 DC rating:
LD,cont = 0.8A @ 25°C

Precharge initial current:
0.8A nominal at 800V
1.14A worst-case with -30% PTC tolerance

Zwei Dioden in Reihe für
Isolation zwischen Fahrzeug
und HVBatt
3*V_MAX_HVBatt -> 3*800 =
2400V
V_RSM 4kV
1.7V V_f @100mA

Precharge logic state
PC open: PCHRG_ACT = GND
PC closed: PCHRG_ACT = 3.3V

K1101 HOLD-TIME / ABSCHALTSEQUENZ
K1101 = KT12-1A-40L-THT
R_coil = 475R
C1103 = 10uF
Worst case: C_min = 8uF (-20%)
V_J = 12V
V_release konservativ = 1.4V
 $\tau_{hold} = R_{coil} * C_{min}$
 $\tau_{hold} = 475R * 8uF = 3.80ms$
 $t_{hold} = -\tau_{hold} * \ln(V_{release} / V_J)$
 $t_{hold} = -3.80ms * \ln(1.4V / 12V)$
 $t_{hold} = 8.16ms$
Abschalten:
1. PCHRG_DELAY_IN / PCHRG_Enable wird LOW.
2. D1106 entlädt den RC-Knoten schnell.
3. U1101 schaltet ab.
4. Q1104 Gate wird ueber R1107 entladen.
5. K1101 bleibt durch C1103 noch ca. 8ms gehalten.
6. K1101 oeffnet erst nach Q1104-OFF.
=> Q1104 ist vor Kontakt-Oeffnung abgeschaltet.
=> Danach bleibt nur der strombegrenzte Pfad ueber
TH1101 + TH1102 + Q1102 bis K1101 oeffnet.

Q1104 PRECHARGE GATE DRIVE / WORST CASE

Bestueckung:
Q1104 = IRLML0100TR
Q1102 = IXTA08N100D2HV
U1101, U1102 = TCLT1004
U1104 = SN74LVC2G14DBV
R1106 = 100R
R1107 = 100k
R1108 = 330R
R1109 = 1k
D1102 + D1104 = ca. 1.7V
D1105 = 10V Gate-Source Clamp

Architektur:
Q1104 brueckt nur Q1102.
TH1101 + TH1102 bleiben immer in Serie.
Q1104 ist daher nicht der Hauptstrom-Bypass wie
bei DCDC_AIR,
sondern reduziert nur den Spannungsabfall / Verlust
an Q1102
nachdem der Precharge-Pfad aktiv ist.

Annahmen Gate-Drive:
+5V_TS_P_VEHl liegt vor D1102/D1104
V_C,U1101 = 5.0V - 1.7V = 3.3V
V_Esat,U1101,max = 0.3V
=> V_GS,Q1104,final,worst = 3.3V - 0.3V = 3.0V

U1101 Eingang:
IF_U1101,min = (3.3V - 1.6V) / 330R = 5.15mA
IF_U1101,typ = (3.3V - 1.25V) / 330R = 6.21mA
IF_limit = 60mA
=> U1101 LED-Strom OK

R1108 Verlustleistung:
P_R1108,typ = (3.3V - 1.25V)^2 / 330R = 12.7mW
=> R1108 thermisch unkritisch

U1101 Ausgangsreserve:
konservativ mit CTR_min = 56%:
IC_U1101,min = 0.56 * 5.15mA = 2.88mA

U1101 statischer Strombedarf bei V_GS = 3.0V:
I_R1107 = 3.0V / 100k = 0.03mA
I_U1101_req,static -= 0.03mA

=> U1101 Reserve statisch > Faktor 90
=> Gate-Endspannung Q1104 worst case ca. 3.0V

Hinweis:
Es gibt keine grosse Gate-Timing-Kapazitaet mehr.
Das Timing wird durch LV-RC + U1104 bestimmt.

Q1104 EINSCHALTEN / LV-DELAY

Bestueckung:
R1116 = 100k
C1107 = 100nF
C1108 = 100nF
C1109 = 100nF
C_sum = 300nF
U1104 = SN74LVC2G14DBV, zwei Inverter in Serie

Wichtig:
U1104 darf nur mit einem 3.3V-kompatiblen Signal
betrieben werden.
PCHRG_DELAY_IN = 3.3V-kompatibles, begrenztes
PCHRG_Enable-Signal.

$\tau_{ON} = R1116 * C_{sum}$
 $\tau_{ON} = 100k * 300nF = 30ms$

Schmitt-Schwelle angenommen:
V_T+ = 1.6...2.3V bei VCC = 3.3V

$t_{ON} = -\tau_{ON} * \ln(1 - V_{T+} / 3.3V)$

Fuer V_T+ = 1.6V:
 $t_{ON} = -30ms * \ln(1 - 1.6V / 3.3V) = 20ms$

Fuer V_T+ = 2.3V:
 $t_{ON} = -30ms * \ln(1 - 2.3V / 3.3V) = 36ms$

=> U1101 / Q1104 wird ca. 20...36ms nach
PCHRG_DELAY_IN eingeschaltet.

Gate-Ladung Q1104 ist gegenueber 20...36ms
vernachlaessigbar.
=> Q1104-ON wird durch
R1116/C1107/C1108/C1109 bestimmt,
nicht durch das MOSFET-Gate.

Q1104 AUSSCHALTEN

Bei PCHRG_DELAY_IN = LOW:
D1106 entlaedt den RC-Knoten schnell.
U1104 schaltet zurueck.
U1101 LED geht aus.
Q1104 Gate entlaedt ueber R1107.

R1107 = 100k
Keine externe Gate-Timing-Kapazitaet vorhanden.

=> Q1104-OFF ist deutlich schneller als
K1101-Release.
=> Q1104 wird vor dem mechanischen Oeffnen von
K1101 abgeschaltet.

Wichtig:
Nach Q1104-OFF bleibt der Precharge-Pfad nicht
hart aus,
sondern laeuft bis K1101 oeffnet weiterhin ueber:
TH1101 + TH1102 + Q1102

Das ist beabsichtigt:
Q1104 ist nur der Q1102-Bypass, nicht der alleinige
Precharge-Schalter.

Lukas Elektronik

Sheet: /Precharge/

File: Precharge.kicad_sch

Title: PackController

Size: A4

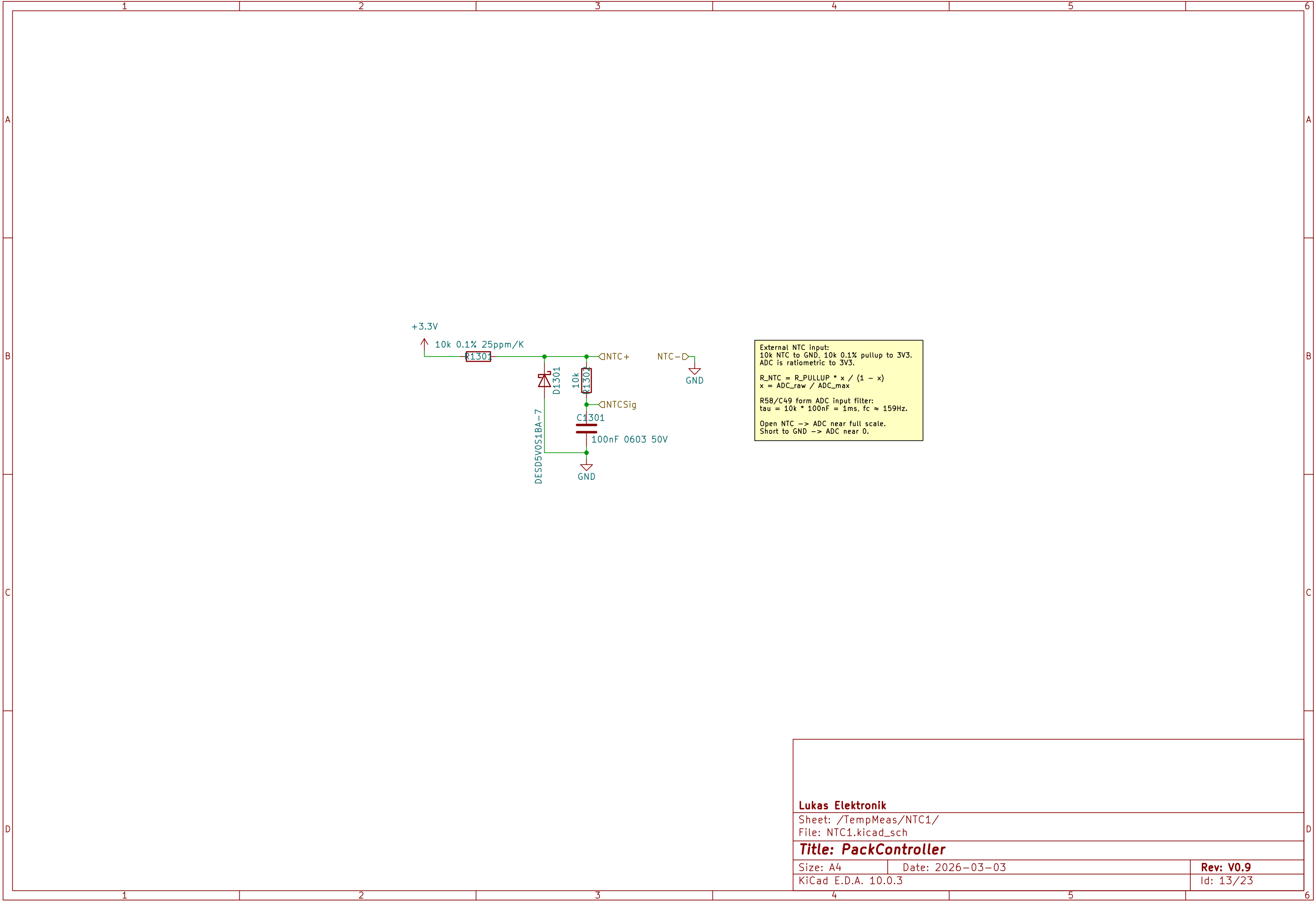
Date: 2026-03-03

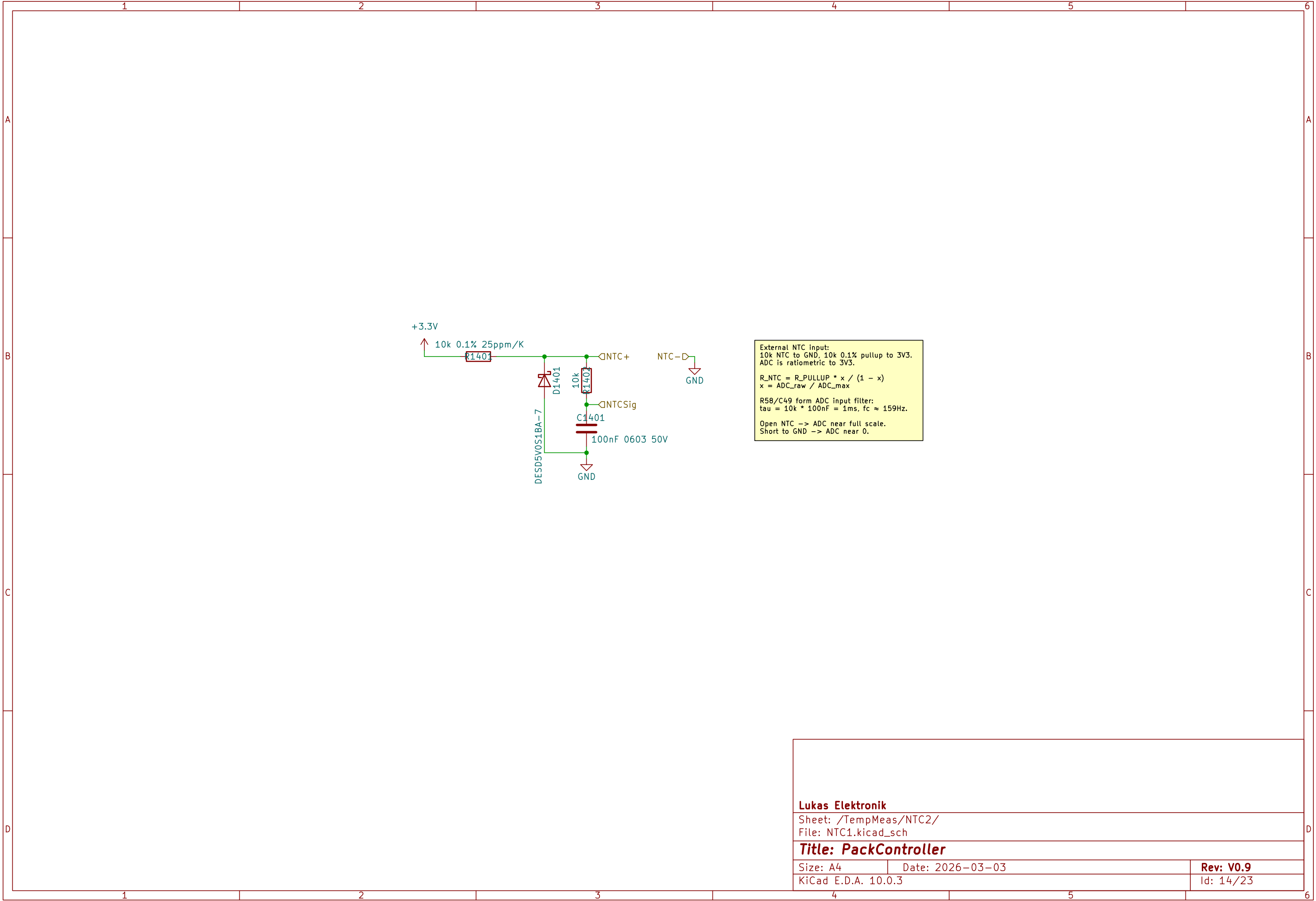
Rev: V0.9

KiCad E.D.A. 10.0.3

Id: 11/23







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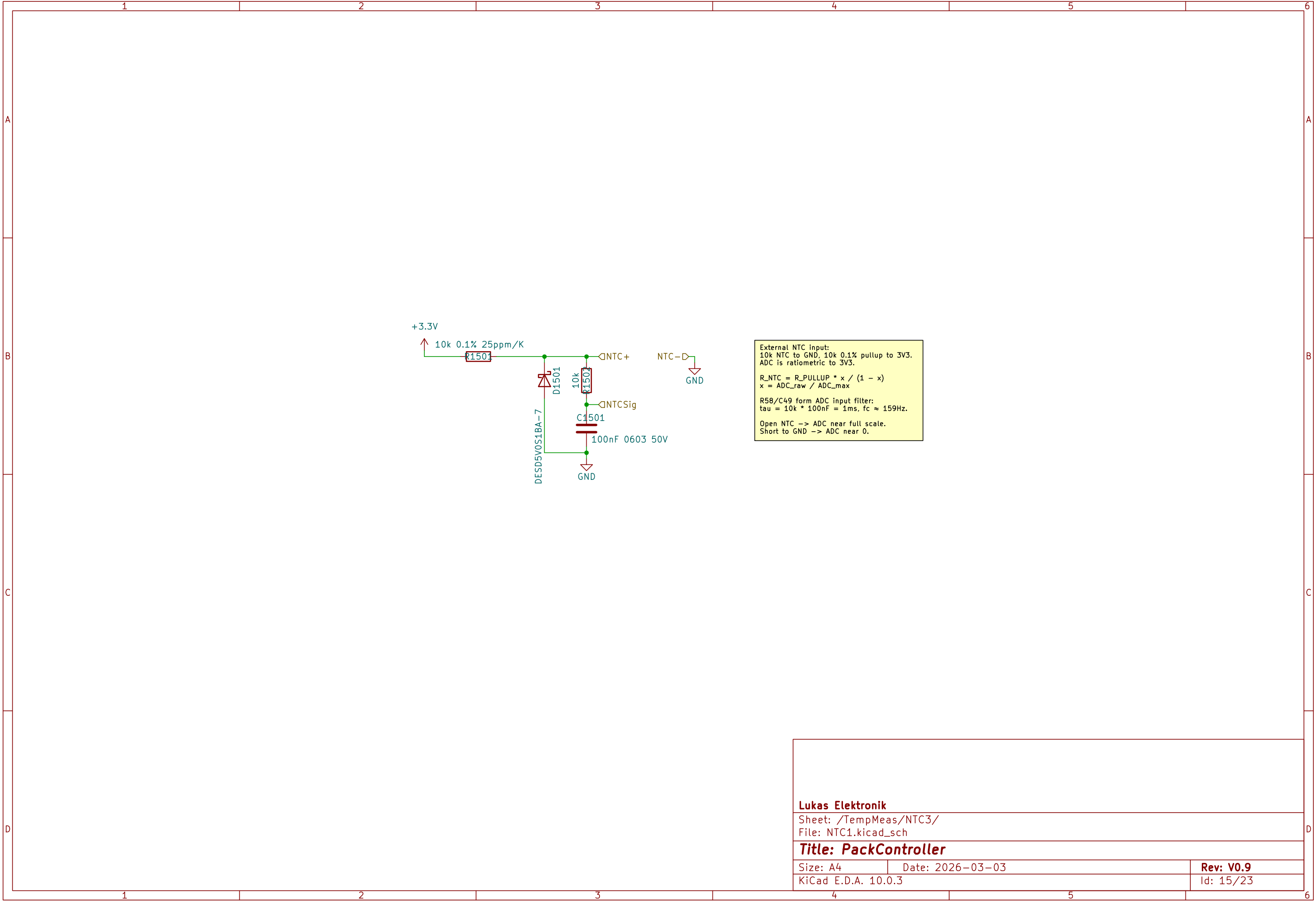
6

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Leakage sensor:
Flex electrodes form R_LEAK. R_EOL = 10M is mounted on the flex sensor, parallel to the electrodes, for open-wire detection.

$R_SENSOR = R_LEAK \parallel R_EOL$

Measurement:
Pulse R_Leak_Supply HIGH, wait 10...50ms, sample ADC, then set supply LOW/Hi-Z. Suggested scan rate: 1Hz.

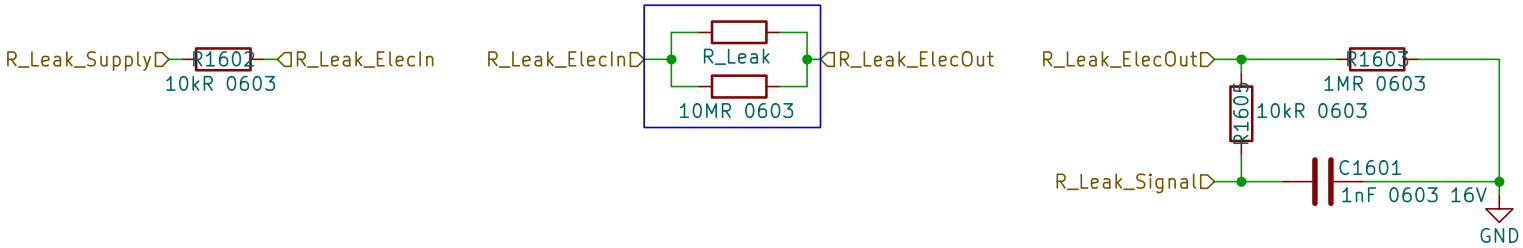
ADC conversion, assuming ADC reference = VDD and R_Leak_Supply = VDD:
 $x = ADC_raw / ADC_max$

$R_SENSOR = R_REF * (1/x - 1) - R_SER_SUPPLY - R_ADC_SER$

Nominal:
R_SER_SUPPLY = 10k
R_REF = 1M
R_ADC_SER = 10k
C_ADC = 1nF
R_EOL = 10M on flex

Expected:
disconnected/open: $ADC \approx 0$
dry connected: $R_SENSOR \approx 10M$
wet/leakage: R_SENSOR decreases

Initial thresholds, validate with real coolant:
warning: $R_SENSOR < 3M$
leak: $R_SENSOR < 1M$
severe leak: $R_SENSOR < 300k$



Lukas Elektronik

Sheet: /HumiditySensing/LeakSens1/
File: LeakSens.kicad_sch

Title: PackController

Size: A4

Date: 2026-03-03

Rev: V0.9

KiCad E.D.A. 10.0.3

Id: 16/23

Leakage sensor:
Flex electrodes form R_LEAK. R_EOL = 10M is mounted on the flex sensor, parallel to the electrodes, for open-wire detection.

$R_SENSOR = R_LEAK \parallel R_EOL$

Measurement:
Pulse R_Leak_Supply HIGH, wait 10...50ms, sample ADC, then set supply LOW/Hi-Z.
Suggested scan rate: 1Hz.

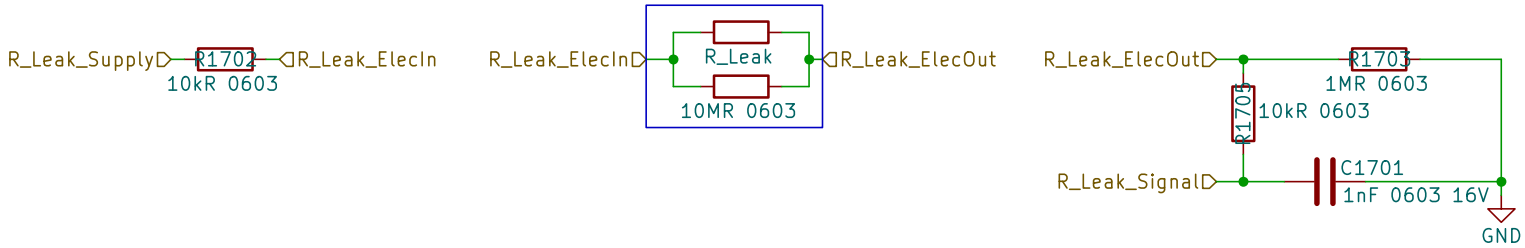
ADC conversion, assuming ADC reference = VDD and R_Leak_Supply = VDD:
 $x = ADC_raw / ADC_max$

$R_SENSOR = R_REF * (1/x - 1) - R_SER_SUPPLY - R_ADC_SER$

Nominal:
R_SER_SUPPLY = 10k
R_REF = 1M
R_ADC_SER = 10k
C_ADC = 1nF
R_EOL = 10M on flex

Expected:
disconnected/open: $ADC \approx 0$
dry connected: $R_SENSOR \approx 10M$
wet/leakage: R_SENSOR decreases

Initial thresholds, validate with real coolant:
warning: $R_SENSOR < 3M$
leak: $R_SENSOR < 1M$
severe leak: $R_SENSOR < 300k$



Lukas Elektronik

Sheet: /HumiditySensing/LeakSens2/
File: LeakSens.kicad_sch

Title: PackController

Size: A4

Date: 2026-03-03

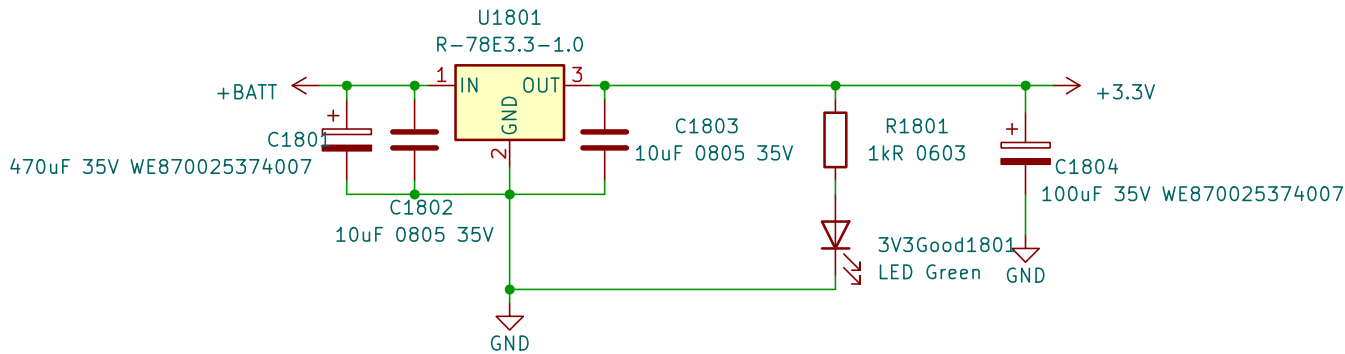
Rev: V0.9

KiCad E.D.A. 10.0.3

Id: 17/23

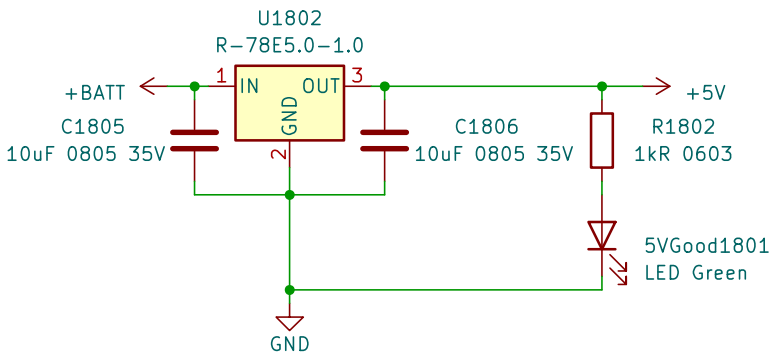
+3V3 Supply worst-case estimate:

Main loads:	
STM32G483 active incl. peripherals:	-50 mA
ADuM3190 VDD1, 2x:	-4 mA
24LC256 EEPROM write:	-3 mA
MCP2562 VIO, 2x worst-case:	-1 mA
ADCM393 comparators + logic:	<1 mA
Pullups, dividers, GPIO drivers, LEDs:	-20...30 mA
Estimated worst-case load:	-80...90 mA
Selected supply:	RECOM 3.3 V / 1 A DC/DC
Design margin:	
1 A / 90 mA $\approx$ 11x	



+5V Supply worst-case estimate:

Main loads:	
TEA1-0505HI isolated DC/DC no-load, 5x:	-120 mA
TEA1 loads incl. ADuM3190 isolated sides:	-25...30 mA
MCP2562 CAN transceivers, 2x dominant:	-140 mA
Misc. reserve / startup / tolerance:	-50 mA
Estimated worst-case load:	-330...400 mA
Selected supply:	RECOM 5 V / 1 A DC/DC
Design margin:	
1 A / 0.4A $\approx$ 2.5x	



Lukas Elektronik

Sheet: /Supply/

File: Supply.kicad_sch

Title: PackController

Size: A4

Date: 2026-03-03

Rev: V0.9

KiCad E.D.A. 10.0.3

Id: 18/23

4.4mm creepage per R
for 2512 inch package

R2004
4.7MR 2512 3kV 0,1%

R2005
4.7MR 2512 3kV 0,1%

R2006
26.1kR 2512 500V 0,1%

TS_N_ACCUD

TS_P_DCDC

+3.3V_DCDC

U2003

1 VDD1

2 INP

3 SNSN

4 GND1

C2004
100pC 0603 50V

C2010
1uC 0805 50V

C2002
100nC 0603 50V

+5V_DCDC

U2004
LP5907MFX-3.3

5 OUT

IN

EN

1

3

2 GND

C2015
100nF 0603 100V

C2016
1uF 0805 50V

C2003
100nF 0603 100V

C2005
10uF 0805 50V

U2002

7 +Vout

5 -Vout

TS_N_ACCUD

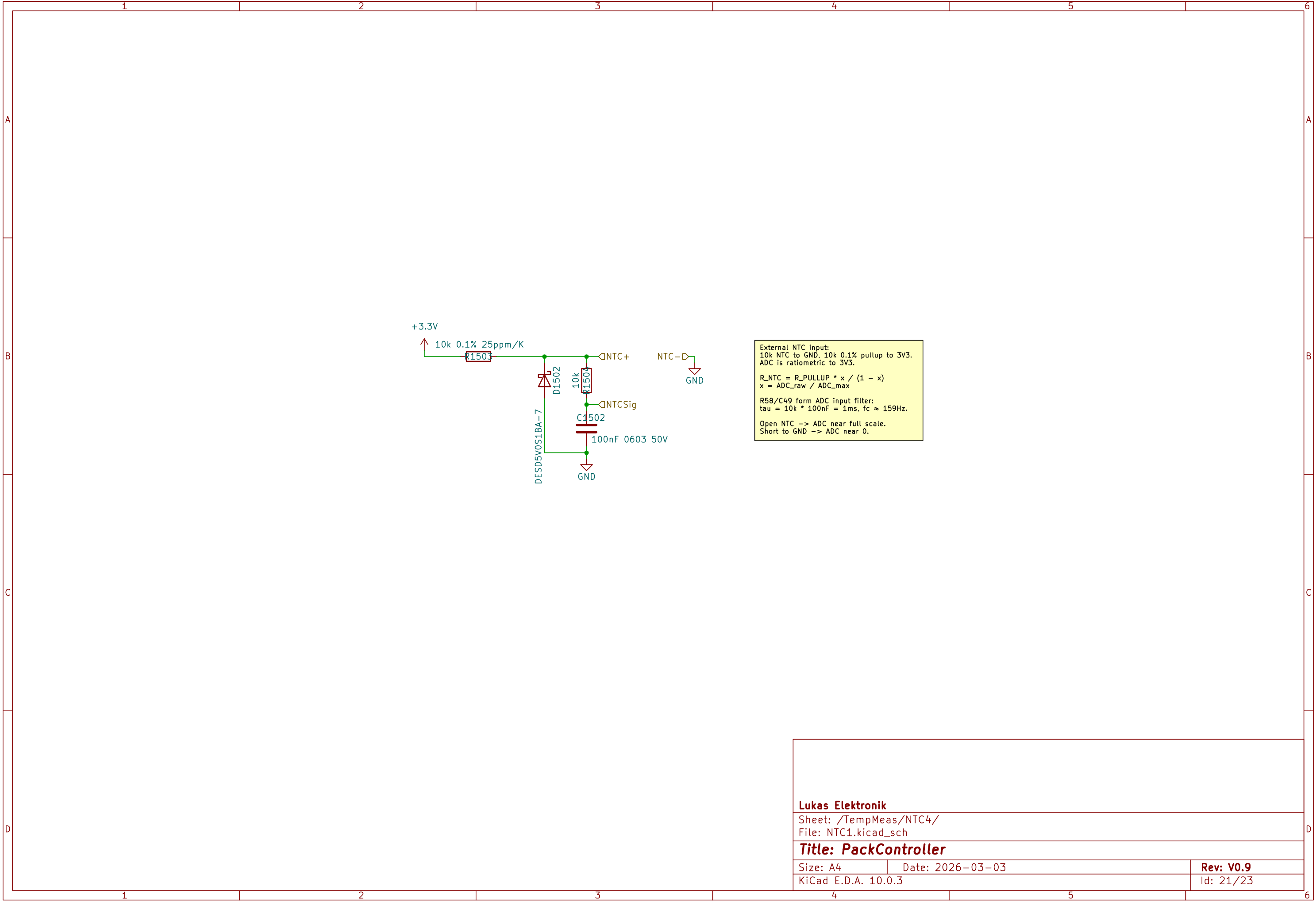
[illegible]

AMC INPUT LOW-PASS

$$R_{TH} = R_{TOP} || R_{SNS}$$
$$R_{TH} = 9.4 \text{ MOhm} || 26.1 \text{ kOhm}$$
$$R_{TH} \text{ approx. } 26.03 \text{ kOhm}$$
$$C_{IN} = C2004 = 100 \text{ pF}$$
$$\tau_{IN} = R_{TH} * C_{IN}$$
$$\tau_{IN} \text{ approx. } 2.60 \text{ us}$$
$$f_{c,IN} = 1 / (2 * \pi * R_{TH} * C_{IN})$$
$$f_{c,IN} \text{ approx. } 61.1 \text{ kHz}$$
$$t_{10-90,IN} \text{ approx. } 2.2 * \tau_{IN}$$
$$t_{10-90,IN} \text{ approx. } 5.7 \text{ us}$$

AMC OUTPUT / ADC LOW-PASS

$$R_{OUT} = R2008 = 1.0 \text{ kOhm}$$
$$C_{OUT} = C2014 = 1.0 \text{ nF}$$
$$\tau_{OUT} = R_{OUT} * C_{OUT}$$
$$\tau_{OUT} = 1.0 \text{ us}$$
$$f_{c,OUT} = 1 / (2 * \pi * R_{OUT} * C_{OUT})$$
$$f_{c,OUT} \text{ approx. } 159 \text{ kHz}$$
$$t_{10-90,OUT} \text{ approx. } 2.2 * \tau_{OUT}$$
$$t_{10-90,OUT} \text{ approx. } 2.2 \text{ us}$$



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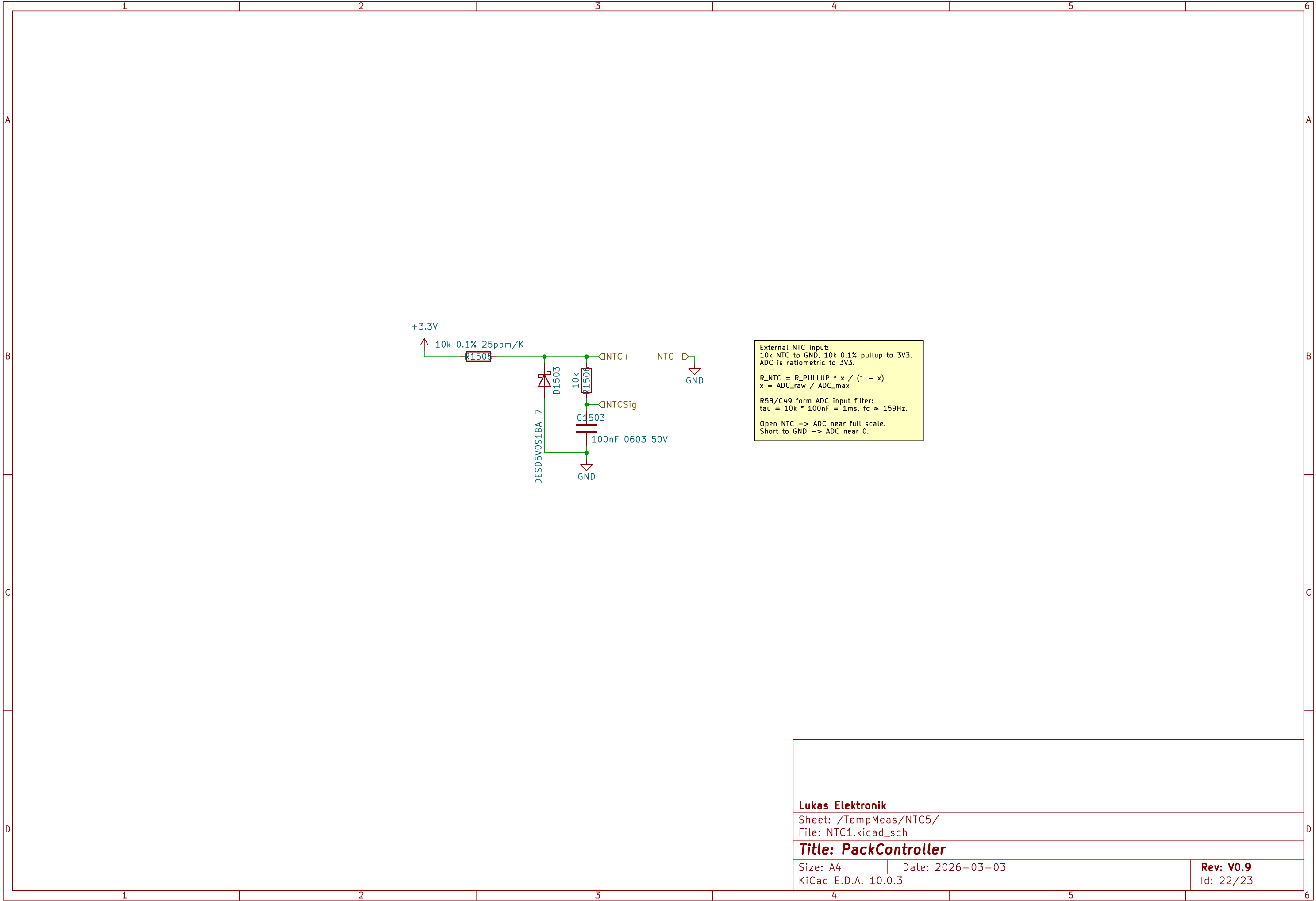
6

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Lukas Elektronik

Sheet: /TempMeas/NTC5/
File: NTC1.kicad_sch

Title: PackController

Size: A4

Date: 2026-03-03

Rev: V0.9

KiCad E.D.A. 10.0.3

Id: 22/23

